

M&M: Molding and Melting Method Using a Replica Diffraction Grating Film and a Laser for Decorating Chocolate with Structural Color

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ABSTRACT

Chocolate is a great food loved around the world. Methods to decorate chocolate with patterns drawn in structural colors have been developed; however, the methods which requires precision molds with nanoscale processing takes a lot of cost and time. In this paper, I propose a new method to decorate chocolate with structural color using a laser engraving machine and a replica diffraction grating film. The proposed method is composed of two simple steps: 1) molding chocolate on a replica diffraction grating film and 2) melting chocolate with structural color with a laser to draw a design. The proposed method allows creation of chocolates decorated with structural color with only a simple manufacturing process and inexpensive equipment without special precision molds.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

human-chocolate interaction, structural color, digital fabrication

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1 INTRODUCTION

The origins of chocolate are very old, usually traced back to the Maya people, who were probably the first to cultivate the cacao plant, in 400 AD [5]. Chocolate that was brought to Spain in the 16th century has evolved from a bitter, spicy drink to tempting foods with various flavors as we know today through innovations.

In addition to chocolatiers' traditional techniques, digital fabrication methods to create novel chocolate, which we call *human-chocolate interaction*, have also been studied [1–4]. For example,

a method of writing messages on chocolate using a laser engraving machine [3] and a three-dimensional (3D) printer to create chocolates with complex structures [2] have been proposed.

Researchers have proposed novel digital fabrication methods to decorate chocolate with structural colors instead of colorants [1, 4]. Unlike colorants that use light absorption and emission, structural color is a phenomenon in which light interferes with microstructures such as grooves and protrusions, causing colors to emerge from them at the wavelength of light. This phenomenon which can be seen on morpho butterfly wings, compact disc surfaces, etc. is characterized by a bright metallic luster that appears to glow and change color in various ways depending on the angle of view.

Since it is difficult to directly make nanoscale microstructures on a surface of chocolate using a laser engraving machine or 3D printer, chocolate with structural color is made by casting melted chocolate in a mold with a nanoscale microstructure. For example, do-it-yourself hobbyists make chocolate with structural color by molding chocolate on an inexpensive replica diffraction grating film (RDGF) with microscopic grooves [1]; however, it cannot create a fine design because the structural color appears over the entire contact surface. Morphotonix has made it possible to produce chocolate with a complex design described in structural colors by adopting a special mold engraved with microstructures [4]. Unfortunately, it is not practical for individuals or small organizations because making molds with microstructures is very difficult and expensive.

I propose a new method that can create a design of structural colors on chocolate using a laser engraving machine and a RDGF. The proposed method mainly consists of two steps: 1) molding chocolate on a RDGF as the conventional method [1] and 2) melting chocolate with structural color with a laser engraving machine to draw a design. Since structural colors are produced by nanoscale microstructures, if the microstructure is disrupted, the color will disappear. The proposed method allows various designs to be drawn on chocolate by melting its microstructure with a laser to create areas with and without structural color.

I consider that the proposed method without expensive molds used in the conventional method [4] will contribute to the development of human-chocolate interaction. For example, it allows anyone to easily fabricate chocolate decorated with structural color with an inexpensive RDGF made of plastic and a low-power laser engraving machine. Moreover, the proposed method can easily and quickly create various designs with laser engraving on pre-made chocolate with structural color. In contrast, a conventional method [4] requires not only an expensive mold for each design but also a long wait for the chocolate to harden; hence, it is difficult to change a design of chocolate immediately. Thus, the proposed method enables a new interaction to create chocolates with customized patterns and messages according to the customer's order at the storefront.

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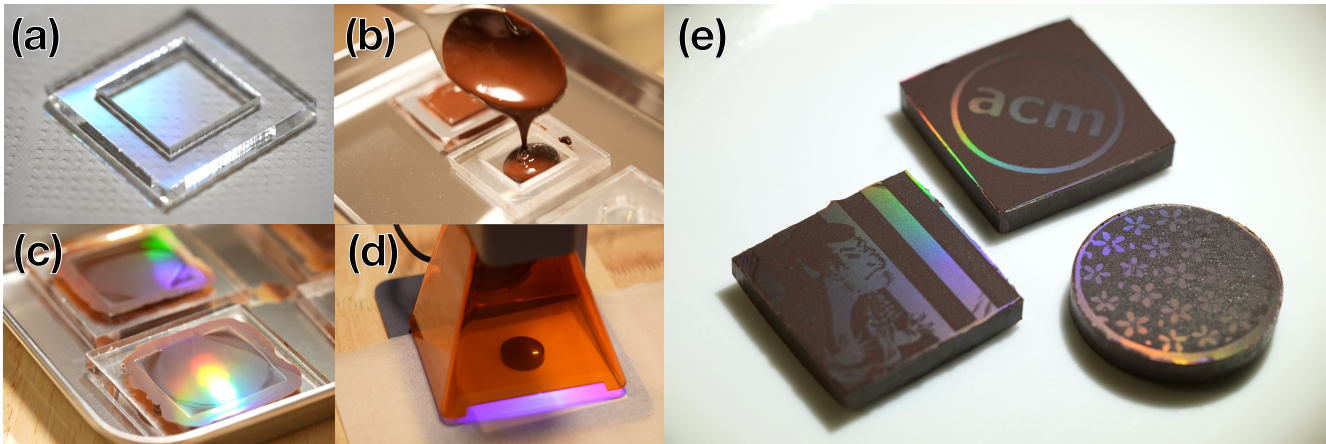


Figure 1: (a) A mold with a RDGF set at the bottom. (b) Pouring tempered chocolate into molds. (c) Chocolate with the diffraction grating removed after solidification. Structural color appears on the surface of the chocolate. (d) Unnecessary structural color area is melted and erased by laser. (e) Structural color chocolates with various decorations.

2 IMPLEMENTATION

I explain the implementation process and tips to decorate chocolate with structural color in this chapter. The chocolate mold shown in Figure 1a is an acrylic frame placed over a RDGF. The RDGF (Edmund Optics, #54-509) is a clear polyester film with 500 grooves per 1 mm. The side with these grooves is the one in contact with the chocolate. Note that if chocolate is cast on the side without the grooves, the structural color does not appear.

Chocolate properties are greatly influenced by the crystal structure of its main ingredient, the fat molecule, triacylglycerol (TAG). The TAGs in this chocolate have crystal structures ranging from Form I to Form VI, each with different properties. The process of changing the crystal structure inside chocolate to the desired one for food by changing temperature is called tempering. The tempering temperature varies depending on the type of chocolate. I tempered couverture dark chocolate (70% cocoa, lubeca) with a low temperature cooker and a stainless-steel bowl as below: 1) warming the chocolate to 50°C until it melts completely; 2) cooling the melted chocolate to 28°C; and 3) heating it to 31°C. The above operations make the crystal structure of Form V chocolate, which is the most suitable for food products because of its smooth, shiny and melt-in-mouth properties. The tempered chocolate is poured into the molds and cooled in a refrigerator at 15°C, as Figure 1b shows.

Here are notes on tempering. First, in each step of tempering, stir the chocolate with a rubber spatula to make it uniform. Second, do not warm the chocolate above 33°C in order not to make the crystal structure of Form VI chocolate after chilling it to 28°C. Third, do not mix the chocolate with water. If water is mixed in, it becomes abnormally hard or does not solidify.

Figure 1c shows the chocolate turned over with its film removed. If tempering and molding processes are carried out correctly, structural color appears on the surface where the film has been applied as Figure 1c shows.

Next, the structural color surface of the chocolate is processed with a laser engraving machine (LaserPecker 2, LaserPecker) as

shown in Figure 1d. The melting point of chocolate is very low at approximately 33°C. Thus, it is more suitable to use inexpensive, low-power models costing several hundred to several thousand dollars using laser-diode, rather than high-power and expensive models using CO₂ or fiber laser. Besides, printing can also be performed at high speeds. We operated a 5W laser engraving machine at minimum output (1%) and maximum speed (100%).

Various patterns and texts can be printed as Figure 1e shows. The floral pattern in Figure 1e is 30 mm in diameter and took approximately 30 seconds to print; hence, a six-pack of chocolates can be printed in about three minutes, which we believe is fast enough for in-store customization for each customer.

I discuss the limitation and future work of the proposed method. Since the proposed method cannot represent gradients, only black and white images are currently available. Besides, we observed the appearance of Form VI crystals in the area irradiated with laser light which was too strong even at the lowest output. This caused a phenomenon called blooming, in which the melted areas turn white over time. I believe that applying a neutral-density filter or reducing irradiation times would improve the issue. In addition, the conventional method using molds [4] is faster than the proposed method for printing large quantities of chocolates with the same design. One way to improve printing speed would be to use a high-powered projector outputting areal light to print rather than scanning a point shape laser. Although the brightness of the structural color is weak with the current prototype, the brightness could be improved by adopting a brazed RDGF that should be more efficient.

3 CONCLUSION

We have presented a new method that can decorate chocolate with structural colors. With the use of a laser engraving machine and a RDGF, the proposed method allows anyone to create structure color chocolate easily at low cost. I have explained and discussed the mechanism, limitation and future work regarding the proposed method.

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